



My computer is more thoughtful than you: Loneliness, anthropomorphism and dehumanization

Hong Im Shin¹ · Juyoung Kim²

Published online: 31 August 2018
© Springer Science+Business Media, LLC, part of Springer Nature 2018

Abstract

Anthropomorphism is the attribution of human traits, emotions and intentions to non-human entities. The inverse process of anthropomorphism is dehumanization in which individuals treat humans like animals or objects. Two studies were conducted to investigate whether there are relationships between loneliness, anthropomorphism, and dehumanization. Study 1 ($N=91$) employed the UCLA loneliness scale in order to examine the extent to which subjective experiences of loneliness related to the tendency to describe nonhuman agents (i.e., animal, computer) with humanlike characteristics. Participants attributed more humanlike capacities to nonhuman agents with increasing levels of self-rated loneliness. Study 2 ($N=70$) induced the emotion of loneliness using an experimental writing task and examined whether there were significant differences in anthropomorphism and dehumanization between the experimental condition of loneliness and the control condition. Results found participants tended to perceive nonhuman agents to be more humanlike in the condition of loneliness than in the control condition. In addition, there was a stronger tendency of dehumanization toward other humans in the condition of loneliness than in the control condition. This implies that anthropomorphism is related to sociality motivation, which is the need for social connection through establishing bonds with nonhumans, and may result in the dehumanization of other humans.

Keywords Loneliness · Anthropomorphism · Social connection · Dehumanization

Introduction

In everyday life, we meet people who treat their pets as humans. They describe their pets not only using attributing behavioral traits (i.e., fast, aggressive), but also humanlike properties such as ‘sympathetic’ and ‘thoughtful’. Therefore, anthropomorphism goes beyond animism, which attributes life to the nonliving. Anthropomorphism

includes attributing humanlike properties and mental states to real or imagined nonhuman agents or objects, while creating inferences about unobservable characteristics of a nonhuman agent (Epley et al. 2007). Thus, to simply describe a nonhuman agent’s behavior as humanlike is not evidence of anthropomorphism. At its core, anthropomorphization entails attributing human mental capacities (i.e., capable of empathy) to nonhuman agents (Epley et al. 2008; Cortes et al. 2005).

Some people perceive animals such as dogs and cats as companion animals for close relationships. There is empirical evidence showing that companion animals provide owners with comparable social support as family and friends (McConnell et al. 2011). The human-animal bond during physical or cognitive pet presence increases self-confidence in goal attainment and reduces blood pressure during distress-eliciting tasks (Zilcha-Mano et al. 2012). Typically, humans are targets of social processes. However, animals can also be seen as social targets (i.e.,

✉ Hong Im Shin
shin7038@naver.com

Juyoung Kim
jkimsg@sogang.ac.kr

¹ School of General Education, Yeungnam University,
Gyeongsan, South Korea

² Sogang Business School, Sogang University, Seoul, South Korea

my cat is my best friend). Sevillano and Fiske (2016) provide an extensive review of animals as social perception targets.¹ They highlighted the bi-directionality of human-animal social perception: anthropomorphism and reverse anthropomorphism. Some animals such as cats were evaluated as positive and perceived as having human mental capacities (anthropomorphism). Conversely, some people associate animal characteristics for out-groups (reverse anthropomorphism, dehumanization), and treat humans like animals or objects. Similar to human social perception, positivity bias (i.e., my good dog, Jack, bit a boy three times and should be forgiven and loved) for ingroup members and discrimination against outgroup members (Haslam and Loughnan 2014) are also relevant in the context of human-animal social perception (Rajecki et al. 2007).

Further, previous research has shown that some individuals attribute humanlike capacities to mechanical agents (i.e., computers have its own independent goals and experienced emotion). According to Leyens et al. (2003), people tend to attribute more secondary emotions (i.e., love, hope) to their ingroup members than to other groups. These results imply that people perceive their ingroup as more human, and consider outgroups as less human. Therefore, people who anthropomorphize more than others are likely to create new bonds with nonhuman agents as ingroup members. In this article, we explore when and why some people anthropomorphize and dehumanize more than others. Who considers nonhumans as humans? Why do some individuals perceive nonhumans to be humanlike? What mechanisms cause some people to view humans as nonhumans?

The Three-Factor Theory of Anthropomorphism: Elicited Agent Knowledge, Sociality Motivation, and Effectance Motivation

According to the three-factor theory (Epley et al. 2007), humans tend to optimize interactions with other social agents. In cases which humans cannot interact with a given social

target through activating sufficient knowledge, they tend to apply highly accessible knowledge as alternatives for making inferences about the agent. There are three primary factors for anthropomorphizing: elicited agent knowledge, sociality motivation, and effectance motivation. These determinants should identify and explain subjects that tend to attribute more humanlike capacities to nonhuman agents, and what circumstances contribute to the stronger tendency of humanizing nonhuman agents. The question here is not the accuracy of anthropomorphic inferences, but individual and situational variabilities in anthropomorphism.

Epley et al. (2007) suggested that the primary cognitive determinant of anthropomorphism is knowledge regarding humans. Such knowledge is highly accessible during inductive reasoning concerning nonhuman agents because it is acquired earlier and more detailed than knowledge involving nonhumans. Inductive reasoning is defined as knowledge transfer from known subjects/objects to novel or unknown subjects/objects (Heit 2000). In general, induction concerning the mental states of animals (i.e., empathy, competence) is based on the perceived similarity (i.e., similar movement, physical similarity), in cases in which no conceptual information is given (Urquiza-Haas and Kotrschal 2015). For example, Morewedge et al. (2007) demonstrated that participants had a stronger tendency to perceive nonhuman targets (i.e., animals, robots) as possessing human mental states when targets moved at speeds similar to natural human movement (i.e., dog, cat). Therefore, inductive inference is prominent in anthropomorphism when little information is available and targets are perceived as similar to human agents. Perceived similarity seems to motivate people to connect with social agents and to perceive them as in-group members. When a social agent appears dissimilar to the self, people are more likely to fail to see humanlike traits and to treat the agent as humanlike (Waytz et al. 2010).

According to Epley et al. (2007), there is less likelihood of anthropomorphizing nonhuman agents in cases in which knowledge about nonhumans is highly accessible. For example, people with highly accessible knowledge regarding nonhumans (i.e., experts about computers) are less likely to anthropomorphize these mechanical agents compared to people with less accessible knowledge. Therefore, the anthropomorphizing tendency will increase for individuals without adequate knowledge concerning nonhuman agents. To our knowledge, no research has been conducted to investigate whether highly accessible knowledge impacts anthropomorphic induction involving nonhuman agents. Therefore, this study attempts to test whether the anthropomorphizing tendency varies depending on students' majors. For example, it is expected that students majoring in computer science are less likely to anthropomorphize computers than students majoring in literature.

According to the three-factor theory of anthropomorphism (Epley et al. 2007), elicited agent knowledge works as a

¹ Sevillano and Fiske (2016) adapted the stereotype content model (SCM; Fiske et al., 2002) to animals to present them as legitimate social targets. The SCM integrates two basic dimensions of social perception: warmth and competence. For example, dogs and cats are seen as warm and competent, whereas rats are perceived as having low warmth and low competence. This may explain why dogs and cats are as ingroup more socially accepted than rats. According to Sevillano and Fiske (2016), animals are special and they are to be differentiated from other inanimate objects, because people see them as acting autonomously, and experiencing emotions. Dogs and cats are perceived as intellectual and friendly. They are considered as companion animals and people provide them with a privileged social position. Compared to this, rats and bugs are perceived as having low intellectual abilities. Consequently, they have socially lower status. When thinking about the context of human targets, dogs and cats are often associated with reference groups (in-group), and rats and bugs with homeless people (out-group).

cognitive mechanism, whereas sociality and effectance are motivational mechanisms. Sociality is the need for social connection with other humans. Lacking social connection may lead people to seek social support. The desire for establishing social relationships might be satisfied through anthropomorphism. For instance, it is hypothesized that people who feel alone are more likely to perceive nonhuman agents as possessing human mental states (i.e., computers have its own mind and feelings). However, effectance motivation describes the need to interact with one's environment effectively. Therefore, lacking certainty might encourage individuals to control their environment through applying knowledge concerning humans to unknown nonhuman agents in order to reduce uncertainty and interact with them effectively. For example, it might be expected that people with little alternative knowledge structure who attempt to understand often-incomprehensible mechanical agents more effectively, are more likely to humanize these mechanical products.

These three determinants are separate variables which work together (Epley et al. 2007). Anthropomorphic knowledge and knowledge concerning humans is activated and applied when effectance and social motivation is increasing. However anthropomorphization decreases if the goal of effectance motivation is fulfilled with an alternative knowledge structure. For example, a student majoring in computer science is unlikely to humanize a computer compared with a novice computer user. However, he or she may humanize a computer when the goal of sociality motivation is not achieved. Effectance should, therefore, increase anthropomorphism when no alternative knowledge structure is available. In contrast, sociality should increase anthropomorphism until sociality motivation is satisfied, even if uncertainty is overcome by an alternate knowledge structure.

Considering these three variables, it can be expected that people humanize other nonhuman agents in order to communicate effectively with limited prior information concerning the agent. However, what happens when social deprivation is also overcome? Conversely, individuals might dehumanize other humans when they feel connected with nonhuman agents. Waytz and Epley (2012) demonstrated that experimentally induced feelings of being socially connected (i.e., writing about family or friends) led people to the increasing tendency of dehumanizing socially distant people (i.e., disabled people, drug addicts). They interpreted these results in terms of the decreasing need for social connection. Therefore, feeling socially connected may promote a disconnection with out-group members'. However these questions remain unanswered: Do feelings associated with nonhuman agents have the same impacts on dehumanization? Do some people dehumanize more than others when they perceive a nonhuman subject as belonging to their in-group?

In this article, we explore these questions. Specifically, we investigate whether there are relationships between

anthropomorphism, loneliness and dehumanization. Until now, limited research (Bartz et al. 2016; Epley et al. 2008) has been conducted to investigate the relationships between anthropomorphism and the underlying psychological mechanisms. Prior to exploring our studies, we will review previous research regarding the association between loneliness, anthropomorphism and dehumanization.

Loneliness, Anthropomorphism and Dehumanization

Loneliness is an unpleasant experience (Hawkley and Cacioppo 2010) due to the human need for social connection (Baumeister and Leary 1995; Epley et al. 2008; Powers et al. 2014). Because loneliness is defined as the subjective experience of feeling alone (Peplau and Perlman 1982; Cacioppo et al. 2015), the physical presence of significant others may be insufficient in preventing feelings of being alone (Cacioppo et al. 2015). This means that people may feel lonely even when being with family and friends. People need to feel connected to significant others in order to prevent feelings of being alone. Peplau and Perlman (1982) defined loneliness as the discrepancy between the preferred state and the actual state of social relations. This discrepancy leads to a subjective experience of loneliness, and motivates people to try to establish social connections (Powers et al. 2014).

The need for social connection is similar to a fundamental drive state like hunger (Baumeister and Leary 1995). People who feel hungry search for food. People who feel socially disconnected attempt to establish social relationships with humans or non-humans (Epley et al. 2007). Feelings of loneliness increase effort and willingness to connect with other non-humans in order to satisfy their need for social connection (Waytz and Epley 2012). Unfortunately, this relationship does not appear so simple. There are examples in everyday life that lonely people treat nonhumans as humanlike, but dehumanize socially distant people such as members of a perceiver's outgroup (i.e., older people, young people, particular stereotyped group) (Waytz et al. 2010). Therefore, the tendency to anthropomorphize nonhuman agents might lead to failure in attributing human characteristics to other people. This suggests that anthropomorphizing and dehumanization may have a shared process; that is, sociality motivation. Feeling disconnected from others may lead to a greater tendency toward anthropomorphism which may satisfy the need for effective interaction with others. Therefore, the present study investigates whether loneliness is related to an increasing tendency of anthropomorphization and dehumanization.

Previous research supports the notion that loneliness might encourage people to connect with one another. For example, feeling disconnected from others leads to the memorization of more social information than feeling connected (Gardner et al.

2000, 2005). Other studies have found that individuals that experienced feelings of isolation expressed a stronger desire for making new friends, and actively searched for connections with others (Maner et al. 2007). In addition, Powers et al. (2014) demonstrated that feeling socially isolated resulted in decreasing thresholds for detecting human faces while discriminating between animate (human) and inanimate (doll) faces. The over-interpretation of animacy implies that feeling alone motivates the employment of adaptive strategies for establishing social relationships. Nonhumans acting as meaningful social agents (i.e., robot, pets) might provide new insights concerning adaptive strategies for coping with psychological difficulties of feeling alone.

Recent research (Bartz et al. 2016; Epley et al. 2008; Waytz and Epley 2012) has demonstrated that lonely people were motivated to maintain social connections and compensated for their lack of social connections through anthropomorphization. Baumeister and Leary (1995) suggested that the need for social connection can be satisfied through various kinds of relationships. These studies support the idea that anthropomorphism can arise from unmet needs for social connection and that people satisfy their belongingness needs through relationships with nonhuman agents (i.e., gadgets, God).

Our present study aims to investigate whether the motivation to fulfill the need for social connections through attributing humanlike qualities (i.e., empathy) to nonhumans (Study 1). The second purpose of this study is to answer the question whether loneliness is related to negative attitudes and dehumanization toward other humans, which is the inverse process of anthropomorphization (Study 2). Previous research (i.e., Cheney and Seyfarth 1990; Harris and Fiske 2006; Hauser 2000; Haslam 2006) has reported the phenomenon of anthropomorphization and dehumanization, but little research (Bartz et al. 2016; Epley et al. 2008; Grunenfeld et al. 2008) has been conducted regarding the psychological mechanisms involved. Therefore, we investigate anthropomorphism based on the three-factor theory (Epley et al. 2007). Specifically, we test whether individuals lacking social connections anthropomorphize more than others during inductive reasoning involving nonhuman agents (e.g., to what extent computers experience its own emotions).

In addition, previous research (Bartz et al. 2016; Epley et al. 2008) has been based on online surveys in which participants could not be effectively monitored. We addressed this potential deficiency by employing a laboratory setting for better quality control. Furthermore, we investigated whether loneliness causes anthropomorphism, and the inverse process, dehumanization, in order to clarify underlying psychological mechanisms. It is expected that increasing loneliness relates to higher scores of anthropomorphism through the active search for compensatory relationships from nonhuman agents. Finally, it is hypothesized that increasing loneliness leads to increasing dehumanization due to the psychological mechanisms of

anthropomorphism have an inverse effect on other humans by treating outgroup members as objects.

Study 1

Method

Participants Ninety-one students (55 female, 36 male; $M_{\text{age}} = 19.9$, age range: 18–27 years) enrolled in introductory psychology courses volunteered for this study. Forty-four students majored in literature (i.e., English literature, German literature), and forty-seven students majored in computer science.

Procedure and Materials Participants completed a twenty-item survey employing a Korean standardized version (Kim 1997) of the UCLA loneliness scale ($\alpha = .91$; i.e., “No one really knows me well” taken from Russell et al. 1980). Our survey used 4-point scales (1: I never feel this way; 2: I rarely feel this way; 3: I sometimes feel this way; 4: I often feel this way). After completing the loneliness scale, participants conducted a task regarding anthropomorphism involving computers and cats (adapted from Epley et al. 2008). The task consisted of two parts. Initially, they had to indicate four anthropomorphic mental-state ratings using 7-point scales ($\alpha = .79$; i.e., the extent to which a computer has “a mind of its own”, has “intentions”, has “free will”, and “experienced emotions.”) and three non-anthropomorphic ratings using 7-point scales regarding computers ($\alpha = .61$; i.e., the extent to which a computer is “attractive”, is “efficient”, and is “strong”). For the second part of the task, participants chose the three best adjectives to describe a cat from a list of fourteen traits. The list included three adjectives related to supportive social relationships (thoughtful, considerate, and sympathetic). Four humanlike words (creative, devious, embarrassing, and jealous) were less related to social connection. The remaining seven words from the list were non-anthropomorphic behavioral descriptions (aggressive, agile, active, energetic, fearful, lethargic, and muscular).

Results and Discussion

The average ratings of loneliness (See Table 1) were significantly correlated to the average anthropomorphic ratings regarding computers ($r(91) = .32, p < .05$). Conversely, the average ratings of loneliness were not significantly correlated to the average non-anthropomorphic ratings regarding computers ($r(91) = .06, p = .63$). Importantly, increased loneliness scores were significantly correlated to an increased proportion of anthropomorphic ratings regarding cats for social-connection traits ($r(91) = .24, p < .05$) concerning the

Table 1 Correlation between loneliness and anthropomorphism regarding computers and cats in Study 1 by the participants ($N = 91$)

	2	3	4	5	6
1. Loneliness	.32*	.06	.24*	.18	-.04
2. Anthropomorphic ratings for computers		.08	-.28*	.06	.01
3. Non-anthropomorphic ratings for computers			-.11	-.15	.05
4. Social-connection traits for cats				-.25*	-.57**
5. Non-connection traits for cats					-.23*
6. Behavioral traits for cats					

Anthropomorphic/non-anthropomorphic ratings for computers are reported as average ratings for each individual item. Four anthropomorphic items for computers included “a mind of its own”, “experienced emotion”, “intention”, and “free will”. Three non-anthropomorphic items for computers were “attractive”, “efficient” and “strong”. The ratings for cats are reported as the proportion of traits selected out of the total number possible for the indicated category (three social connection traits, four non-connection traits, and seven behavioral traits). * $p < .05$, ** $p < .01$

proportion of selected traits calculated for each trait category (three social-connection traits, four non-connection traits, seven behavioral traits). Conversely, there were no significant relationships between loneliness and the proportion of other trait types concerning cats. Loneliness was not related to the proportion of non-connection traits ($r(91) = .18$, $p = .12$) or the proportion of behavioral traits ($r(91) = -.04$, $p = .73$) selected by participants.

When comparing participants by major, there were significant differences between participants majoring in computer science and in literature concerning their average self-rated loneliness scores (See Table 2). Participants majoring in computer science felt increased loneliness compared with those majoring in literature ($t(89) = -2.27$, $p < .05$). Importantly, participants majoring in computer science were more likely to anthropomorphize computers than students majoring in literature ($t(89) = -2.50$, $p < .05$). This tendency extended to anthropomorphic ratings for cats. Participants majoring in computer science selected more traits for social connection regarding cats than those majoring in literature ($t(89) = -2.05$, $p < .05$). However, there were no other significant differences between participants in choosing non-connection traits and behavioral traits. These results demonstrate that the

extent of loneliness was related to the anthropomorphism of participants regarding computers and cats.

In Study 1, there were significant relationships between self-reported loneliness and the anthropomorphizing tendency. Consistent with our expectations and previous research (Epley et al. 2008), increasing subjective loneliness was correlated with increasing use of humanlike attributes for social connections with computers and cats. Importantly, non-anthropomorphic attributes and non-connection trait words were not related to the average rating of loneliness. This finding implies that the subjective experience of loneliness motivates individuals to seek social connections not only regarding humans, but also pets and computers. However, Study 1 investigated only the relationship between loneliness and the anthropomorphizing tendency, not the cause for the use of humanlike social-connection words. To clarify this interpretation, loneliness was experimentally induced in Study 2. The increasing anthropomorphization in the condition of loneliness might demonstrate that loneliness impacts the selection of trait words for non-human agents for promoting social connection. In addition, Study 2 investigated whether experimentally induced loneliness was related to negative attitudes toward other human

Table 2 Mean and standard deviation (in parentheses) for loneliness and anthropomorphic ratings for computers and cats in Study 1

	Majors of the participants ($N = 91$)		
	Literature ($n = 44$)	Computer Science ($n = 47$)	t, p
Loneliness ratings	1.65 (.31)	1.89 (.47)	$t(89) = -2.27^*$, $p < .05$
Anthropomorphic ratings for computers	2.27 (1.56)	3.21 (1.28)	$t(89) = -2.50^*$, $p < .05$
Social-connection traits for cats	.04 (.11)	.15 (.31)	$t(89) = -2.05^*$, $p < .05$
Non-connection traits for cats	.26 (.19)	.15 (.16)	$t(89) = 2.25^*$, $p < .05$
Behavioral traits for cats	.25 (.12)	.24 (.14)	$t(89) = .48$, $p = .63$

Anthropomorphic ratings regarding computers are reported as the average ratings for the four anthropomorphic items. Non-anthropomorphic ratings regarding computers are reported as the average ratings for the three non-anthropomorphic items. Anthropomorphic ratings regarding cats are reported as the proportion of traits selected out of the total number possible for the indicated category. * $p < .05$

agents. Previous research (Cacioppo et al. 2015) has suggested that lonely people did not seem to seek human agents due to having predisposed negative thoughts concerning others. Therefore, Study 2 compared the condition of loneliness and the control condition for anthropomorphization and dehumanization (i.e., to treat humans as nonhumans, such as animals).

Study 2

Method

Participants Seventy students (39 female; $M_{\text{age}} = 23.5$, age range: 19–25 years) voluntarily participated in Study 2. All participants majored in the social sciences (i.e., psychology, sociology, and economics).

Procedure and Materials Initially, participants were randomly assigned to either the loneliness group or the control group. To experimentally induce the emotion of loneliness, an indirect priming procedure consisting of a writing task was utilized. Previous research (Campbell et al. 2017; Waytz and Epley 2012) has shown that a writing task focused on a specific experience can activate related thoughts and emotions. Concerning the loneliness group ($n = 36$), participants described their experiences of feeling alone during the past semester. Conversely, participants in the control group ($n = 34$) wrote about their activities during the past semester. The writing task was limited to five minutes for both groups. Subsequently, participants completed the same loneliness scale as in Study 1 using a Korean Version ($\alpha = .71$). After completing the loneliness scale, they conducted the identical task of anthropomorphization regarding computers and cats as in Study 1. In Study 2, the Cronbach's alpha (α) for the

anthropomorphic items regarding computers was $\alpha = .62$, and that for the non-anthropomorphic items was $\alpha = .62$. Finally, they completed the dehumanization scale (Waytz and Epley 2012; adapted from Bandura et al. (1996)'s moral engagement scale) in Korean using seven-point scales (1 = Strongly Disagree, 7 = Strongly Agree), which consisted of four items ($\alpha = .63$; i.e., "It is okay to treat badly somebody who behaved like a 'worm'").

Results and Discussion

We predicted that participants in the loneliness group would have a stronger tendency for anthropomorphizing computers and cats than those in the control group. In addition, we hypothesized that participants would exhibit greater dehumanization in the loneliness group than those in the control group if the underlying psychological mechanisms for anthropomorphization is associated with dehumanization. We present the analyses of the individual measures and then the overall analysis across all measures.

Manipulation Check

As shown in Table 3, there were significant differences between the loneliness group and the control group regarding the self-reported loneliness scores ($t(68) = 2.13, p < .05$). Experimental manipulation induced stronger feelings of loneliness in the loneliness group compared with the control group.

Anthropomorphization

As predicted, there was a stronger tendency to attribute human-like traits to computers in the loneliness group than in the control group ($t(68) = 2.03, p < .05$). However, there were no

Table 3 Mean and standard deviation (in parentheses) for dehumanization and anthropomorphization regarding computer and cats in the condition of loneliness ($n = 36$) and the control condition ($n = 34$)

	Condition		
	Loneliness	Control	<i>t, p</i>
Loneliness ratings	2.11 (.74)	1.77 (.56)	$t(68) = 2.13^*, p < .05$
Anthropomorphic ratings for computers	2.46 (1.10)	1.92 (.79)	$t(68) = 2.28^*, p < .05$
Social-connection traits for cats	.07 (.18)	.06 (.15)	$t(68) = .13, p = .89$
Non-connection trait for cats	.23 (.20)	.25 (.18)	$t(68) = -.49, p = .68$
Behavioral traits for cats	.66 (.26)	.64 (.25)	$t(68) = .31, p = .75$
Dehumanization ratings	4.09 (1.26)	3.52 (.94)	$t(68) = 2.09^*, p < .05$

As in Study 1, anthropomorphic ratings regarding computers are reported as the average ratings of the four anthropomorphic items. Anthropomorphic ratings regarding cats are reported as the proportion of traits selected out of the total number for the indicated category. Dehumanization is reported as the average ratings of the four items ($\alpha = .74$) adapted from Waytz and Epley (2012) as follows: "Some people deserve to be treated like animals," "It is okay to treat badly somebody who behaved like a 'worm'," "Someone who is obnoxious does not deserve to be treated like a human being," and "Some people have to be treated roughly because they lack feelings that can be hurt" (1 = Strongly Disagree, 7 = Strongly Agree; $\alpha = .63$). * $p < .05$

significant differences between the two groups regarding nonanthropomorphic ratings for computers ($t(68) = -.15$, $p = .88$). The anthropomorphic ratings for cats in the loneliness group were not significantly higher than in the control group ($t(68) = .13$, $p = .89$) (See Table 3). This was unexpected and will be further explored in the discussion section. The higher anthropomorphic ratings for computers suggest that individuals experiencing feelings of loneliness may promote attributing human-like qualities to nonhuman agents to satisfy the need for social connection.

Dehumanization

There were significant differences between the two groups regarding dehumanization. Dehumanization scores in the loneliness group were significantly higher than in the control group, $t(68) = 2.85$, $p < .05$.

Additionally, as expected, increasing scores in self-reported loneliness were associated with tendencies for anthropomorphization and dehumanization (See Table 4). Consistent with Study 1, loneliness was positively related to the anthropomorphic attribution regarding computers ($r(68) = .32$, $p < .01$). In addition, loneliness was positively associated with dehumanization ($r(68) = .24$, $p < .05$). Importantly, increasing anthropomorphization regarding computers was significantly associated with increasing dehumanization ($r(68) = .25$, $p < .05$). It should also be noted that the anthropomorphization regarding cats (selecting social-connection traits) was positively associated with dehumanization ($r(68) = .21$, $p = .07$). These results demonstrate that the underlying psychological mechanisms for anthropomorphization and dehumanization may be identical.

Overall Analysis

To test our predictions, we standardized participants' responses across our three critical dependent measures: anthropomorphic ratings for computers, percentage of social-connection traits selected for cats, and dehumanization scores. A 2 (condition: loneliness, control) $\times$ 3 (measure: anthropomorphic ratings for computers, social-connection traits for cats, dehumanization) mixed-model ANOVA revealed the main effect of condition was significant ($F(1, 68) = 12.11$,

$p < .01$, $\eta^2 = .15$). The standardized scores of anthropomorphic ratings (for computers and cats) and the standardized dehumanization scores were significantly higher in the loneliness condition ($M = .18$, $SD = .08$) than in the control condition ($M = -.17$, $SD = .08$). However, the main effect of measure was not significant ($F < 1$). The interaction effects between condition and measure were not significant ($F(1, 68) = 1.43$, $p = .23$, $\eta^2 = .02$). Therefore the loneliness condition seemed to impact the anthropomorphic ratings for computers and the tendency for dehumanization.

Study 2 investigated whether the experimentally induced experience of loneliness impacts the tendencies for anthropomorphization and dehumanization. Results found there were significant differences between the loneliness condition and the control condition. The attribution of human capacities to computers was more prevalent in the condition of loneliness than in the control condition. In addition, the tendency of dehumanization was higher in the condition of loneliness than in the control condition. However, there were no significant differences between the condition of loneliness and the control condition in the selection of social-connection traits regarding cats. It remains unclear why significant differences in anthropomorphization regarding the evaluation of cats were not evident in Study 2. Other uncontrolled factors (i.e., previous experiences with cats, pet ownership) might impact the anthropomorphic attribution for cats. However, a potential effect of the anthropomorphization of cats cannot be excluded, because the selection of traits for social connection was positively related to the tendency of dehumanization.

General Discussion

In this article, two studies demonstrated that loneliness may be related to anthropomorphization and dehumanization. Study 1 investigated whether increasing loneliness is associated with increasing anthropomorphic ratings regarding computers (i.e., computers have its own emotion) and pets (i.e., cats are considerate). Participants with higher scores in self-rated loneliness were more likely to anthropomorphize nonhuman agents. In addition, participants majoring in computer science tended to

Table 4 Correlation between loneliness, anthropomorphism and dehumanization in Study 2

	2	3	4	5	6
1. Loneliness ratings	.32**	-.05	-.17	.15	.24*
2. Anthropomorphic ratings for computers		.07	-.05	-.03	.25*
3. Social-connection traits for cats			-.14	-.31*	.21
4. Non-connection traits for cats				-.32*	-.08
5. Behavioral traits for cats					.16
6. Dehumanization					

* $p < .05$, ** $p < .001$

feel more alone and were more likely to anthropomorphize computers and cats than other participants. Study 2 demonstrated that participants in the loneliness group attributed more humanlike attributes to computers, and dehumanized other humans. These results imply that social disconnection might cause increased motivation to interact with nonhumans and decreased effort and willingness to connect with other humans. These results support the three-factor theory (Epley et al. 2007), which suggested that anthropomorphism can be a motivational search for social connection, and previous research (Bartz et al. 2016; Epley et al. 2008).

Furthermore, we showed that experiencing loneliness could increase the process of dehumanization. These findings imply that the need for social connection as a psychological mechanism can influence anthropomorphization and dehumanization concurrently. Therefore, lonely people seem to satisfy their need to interact with nonhuman agents through anthropomorphization rather than through connecting with other humans from outgroups. In line with previous research (Cacioppo et al. 2015), which identified negative attitudes toward humans as a critical cause for chronic loneliness, our findings might be interpreted that not loneliness in itself, but negative attitudes of lonely individuals toward humans can account for the stronger tendency for anthropomorphism and dehumanization. Thus, people feeling lonely with negative attitudes toward others do not seek social closeness to other human agents. Rather, it might lead to searching for nonhuman agents for social satisfaction. The strong connections with nonhuman agents may diminish the motivation to interact with other humans. Indeed, our results support the theory that sociality has an inverse effect on anthropomorphism and dehumanization (Epley et al. 2007). A person feeling socially unconnected seeks out humanlike agents. Consequently, feeling socially connected with other nonhuman agents may increase the tendency to perceive other people as socially more distant out-group members. Understanding sociality motivation may answer the question why chronically lonely people are unlikely to establish relationships with other humans. Stronger connections with nonhuman agents may decrease the baseline of sociality motivation and may result in reduced need to establish social bonds with other human agents.

According to the three-factor theory of anthropomorphism (Epley et al. 2007), three primary factors play important roles in perceiving nonhumans as humans: highly accessible knowledge, sociality and effectance. According to our results, sociality seemed most critical for anthropomorphism compared with accessible knowledge and effectance. In Study 1, inconsistent with the theory that people possessing highly accessible knowledge are less likely to anthropomorphize, participants majoring in computer science tended to display greater anthropomorphization regarding computers than those majoring in literature, in spite of having richer cognitive representation of computers. In Study 1, the self-rated scores for loneliness of students majoring

in computer science were higher than those of students majoring in literature. Therefore, the increase in anthropomorphization of students might arise from loneliness rather than from highly accessible knowledge. In future studies, it is necessary to investigate whether the anthropomorphization tendency varies with the extent of accessible knowledge. Recently, Bartz et al. (2016) reported that attachment anxiety (i.e., intense desire for closeness) was a stronger predictor for anthropomorphization than loneliness. It seems noteworthy that anthropomorphization arises from different kinds of unmet needs for social connection, and people form relationships with nonhuman agents to satisfy their belongingness needs. In future studies, it is necessary to investigate whether the tendency of anthropomorphization varies with the extent of sociality motivation.

One limitation of our research is that the association between anthropomorphization concerning computers and loneliness was not observed in the relationships between anthropomorphization of cats and loneliness in Study 2. The presentation order of anthropomorphic items was fixed and not counterbalanced. Therefore, this might impact the anthropomorphic ratings for cats. Additionally, we recruited university students who might not have pets. Previous experience regarding pets was not assessed. Future studies should include additional assessments to investigate the relationship between anthropomorphization concerning pets and loneliness. The second limitation is that we did not measure dispositional loneliness of individual participants. The possibility cannot be excluded that experimentally induced loneliness was confounded with dispositional loneliness. Finally, implicit loneliness was not assessed. For individuals incapable of accurate reporting, or unwilling to report accurately because of biased self-representation, implicit measurements can be informative (Marroquin et al. 2016). In future studies, various assessment tools for loneliness might help to detect the effects of anthropomorphization and dehumanization.

Acknowledgments This work was supported by the Ministry of Education of the Republic of Korea and the National Research Foundation of Korea (NRF-2017S1A5A8018952).

Compliance with Ethical Standards

Conflict Interests The authors declare that they have no conflict of interest with respect to their authorship or the publication of this article.

References

- Bandura, A., Barbaranelli, C., Caprara, G., & Pastorelli, C. (1996). Mechanisms of moral disengagement in the exercise of moral agency. *Journal of Personality and Social Psychology*, 71, 364–374.
- Bartz, J. A., Tchalova, K., & Fenerci, C. (2016). Reminders of social connection can attenuate anthropomorphism: A replication and

- extension of Epley, Akalis, Waytz, and Cacioppo (2008). *Psychological Science*, 27, 1644–1650.
- Baumeister, R. F., & Leary, M. R. (1995). The need to belong: Desire for interpersonal attachments as a fundamental human motivation. *Psychological Bulletin*, 117, 497–529.
- Cacioppo, S., Grippo, A. J., London, S., Goossens, I., & Cacioppo, J. T. (2015). Loneliness: Clinical import and interventions. *Perspectives on Psychological Science*, 10, 238–249.
- Campbell, K. L., Benoit, R. G., & Schacter, D. L. (2017). Priming, not inhibition, of related concepts during future imagining. *Memory*, 25, 1235–1245.
- Cheney, D., & Seyfarth, R. (1990). *How monkeys see the world*. Chicago: University of Chicago Press.
- Cortes, B.P., Demoulin, S., Rodriguez, R.T., Rodriguez, A.P., & Leyens, J.P. (2005). Infrahumanization or familiarity?: Attribution of uniquely human emotions to the self, the ingroup, and the outgroup. *Personality and Social Psychology Bulletin*, 31, 243–253.
- Epley, N., Waytz, A., & Cacioppo, J. T. (2007). On seeing human: A three-factor theory of anthropomorphism. *Psychological Review*, 114, 864–886.
- Epley, N., Akalis, S., Waytz, A., & Cacioppo, J. T. (2008). Creating social connection through inferential reproduction: Loneliness and perceived agency in gadgets, gods, and greyhounds. *Psychological Science*, 19, 114–120.
- Fiske, S. T., Cuddy, A. J. C., Glick, P., & Xu, J. (2002). A model of (often mixed) stereotype content: Competence and warmth respectively follow from perceived status and competition. *Journal of Personality and Social Psychology*, 82, 878–902.
- Gardner, W. L., Pickett, C. L., & Brewer, M. B. (2000). Social exclusion and selective memory: How the need to belong affects memory for social information. *Personality and Social Psychology Bulletin*, 26, 486–496.
- Gardner, W. L., Pickett, C. L., Jefferis, V., & Knowles, M. (2005). On the outside looking in: Loneliness and social monitoring. *Personality and Social Psychology Bulletin*, 31, 1549–1560.
- Grunenfeld, D. H., Inesi, E. M., Magee, J. C., & Galinsky, A. D. (2008). Power and the objectification of social targets. *Journal of Personality and Social Psychology*, 95, 111–127.
- Harris, L. T., & Fiske, S. T. (2006). Dehumanizing the lowest of the low. *Psychological Science*, 17, 847–853.
- Haslam, N. (2006). Dehumanization: An integrated review. *Personality and Social Psychology Review*, 10, 252–264.
- Haslam, N., & Loughnan, S. (2014). Dehumanization and infrahumanization. *Annual Review of Psychology*, 65, 399–423.
- Hauser, M. D. (2000). *Wild minds: What animals really think*. New York: Holt.
- Hawkey, L. C., & Cacioppo, J. T. (2010). Loneliness matters: A theoretical and empirical review of consequences and mechanisms. *Annals of Behavioral Medicine*, 40, 218–227.
- Heit, E. (2000). Properties of inductive reasoning. *Psychonomic Bulletin & Review*, 7, 569–592.
- Kim, O. (1997). Korean version of the revised UCLA loneliness scale: Reliability and validity test. *Journal of Korean Academy of Nursing*, 27, 871–879.
- Leyens, J. P., Cortes, B. P., Demoulin, S., Dovidio, J., Fiske, S. T., & Gaunt, R. (2003). Emotional prejudice, essentialism, and nationalism. *European Journal of Social Psychology*, 33, 703–717.
- Maner, J. K., DeWall, C. N., Baumeister, R. F., & Schaller, M. (2007). Does social exclusion motivate interpersonal reconnection? Resolving the “porcupine problem.”. *Journal of Personality and Social Psychology*, 92, 42–55.
- Marroquin, B., Czamanski-Cohen, J., Weihs, K. L., & Stanton, A. L. (2016). Implicit loneliness, emotion regulation, and depressive symptoms in breast cancer survivors. *Journal of Behavioral Medicine*, 39, 832–844.
- McConnell, A. R., Brown, C. M., Shoda, T. M., Stayton, L. E., & Martin, C. E. (2011). Friends with benefits: On the positive consequences of pet ownership. *Journal of Personality and Social Psychology*, 101, 1239–1252.
- Morewedge, C. K., Preston, J., & Wegner, D. M. (2007). Timescale bias in the attribution of mind. *Journal of Personality and Social Psychology*, 93, 1–11.
- Peplau, L. A., & Perlman, D. (1982). Perspectives on loneliness. In L. A. Peplau & D. Perlman (Eds.), *Loneliness: A source book of current theory, research and therapy* (pp. 1–8). New York: Wiley.
- Powers, K. E., Worsham, A. L., Freeman, J. B., Wheatley, T., & Heatherton, T. F. (2014). Social connection modulates perceptions of animacy. *Psychological Science*, 25, 1943–1948.
- Rajecki, D. W., Ramussen, J. L., & Conner, T. J. (2007). Punish and forgive: Causal attribution and positivity bias in response to cat and dog misbehavior. *Society and Animals*, 15, 311–328.
- Russell, D., Peplau, L. A., & Cutrona, C. E. (1980). The revised UCLA loneliness scale: Concurrent and discriminant validity evidence. *Journal of Personality and Social Psychology*, 39, 472–480.
- Sevillano, V., & Fiske, S. T. (2016). Animals as social objects. *European Psychologist*, 21, 206–217.
- Urquiza-Haas, E. G., & Kotrschal, K. (2015). The mind behind anthropomorphic thinking: Attribution of mental states to other species. *Animal Behavior*, 109, 167–176.
- Waytz, A., & Epley, N. (2012). Social connection enables dehumanization. *Journal of Experimental Social Psychology*, 48, 70–76.
- Waytz, A., Epley, N., & Cacioppo, J. T. (2010). Social cognition unbound: Insights into anthropomorphism and dehumanization. *Current Directions in Psychological Science*, 19, 58–62.
- Zilcha-Mano, S., Mikulincer, M., & Shaver, P. R. (2012). Pets as safe havens and secure bases: The moderating role of pet attachment orientations. *Journal of Research in Personality*, 46, 571–580.